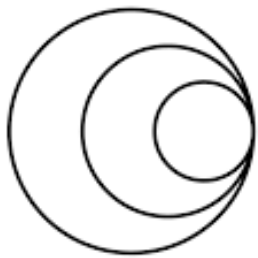


Encapsulated bacteria practicals



**wellcome
connecting
science**

17th September
2025

Haemophilus influenzae, *Streptococcus pneumoniae* and *Neisseria meningitidis* are very different bacterial species

Non-culture diagnosis is important for investigating disease caused by these organisms

Identifying species and the capsule expressed is essential

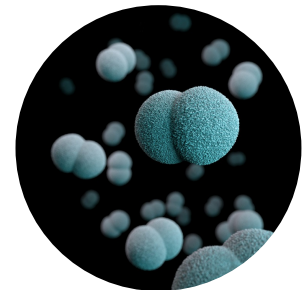
H. influenzae Gram negative rod



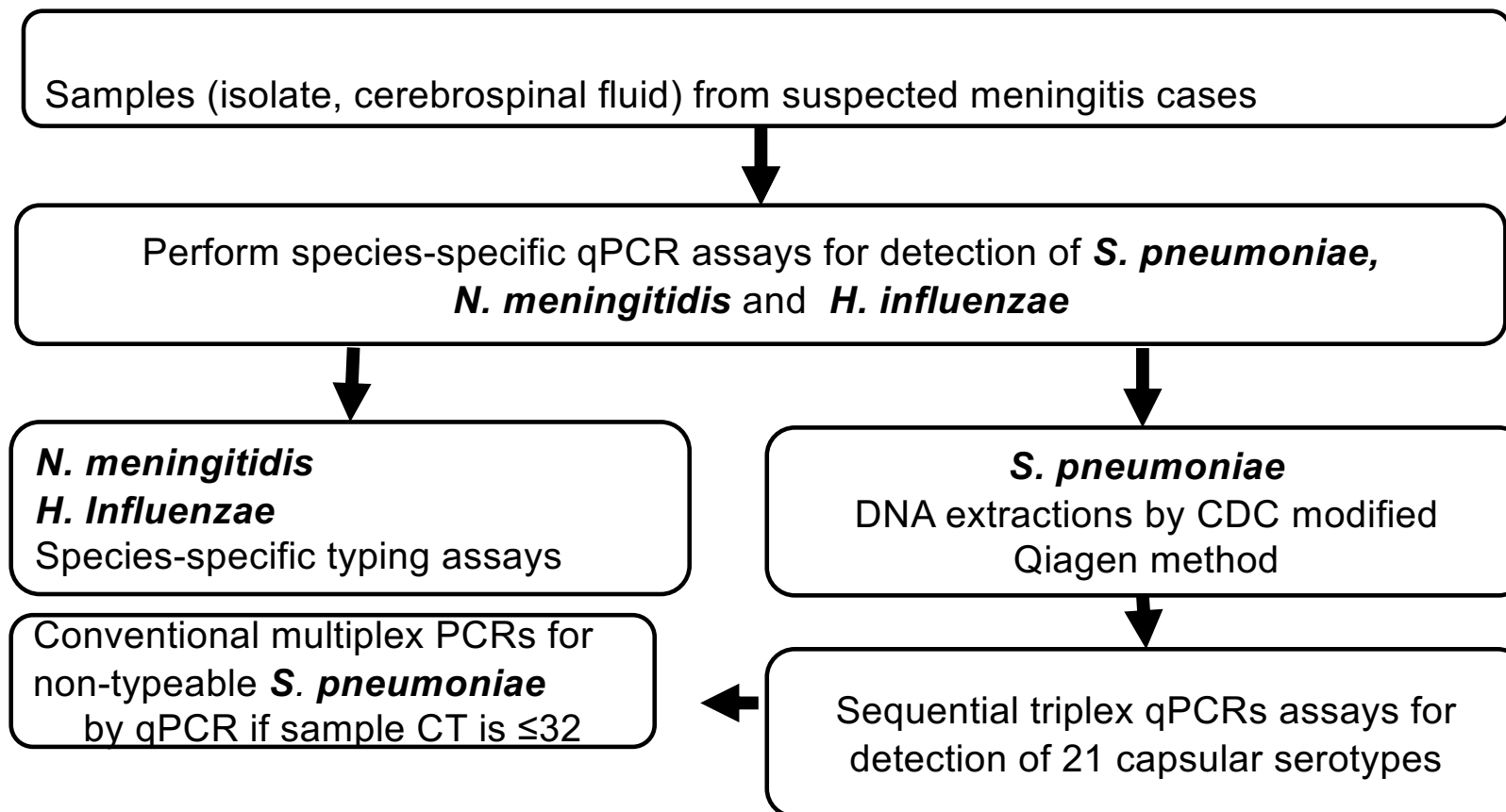
S. pneumoniae Gram positive cocci



N. meningitidis Gram negative diplococci



Workflow for detection and serotyping of bacterial meningitis pathogens by real time PCR



Practical 1 – rt-PCR species identification

		Per group
	Rxn x 1 (μl)	<i>Triplex(lytA-sodC-hpd)</i>
Total needed		5
Mmix	12.5	62.5
H2O	1.5	7.5
lytA-F	1	5
lytA-R	1	5
lytA-P	1	5
sodC-F	1	5
sodC-R	1	5
sodC-P	1	5
hpd3-F	1	5
hpd3-R	1	5
hpd3-P	1	5
Total mix	23	115
CSF/DNA	2	

Cycling conditions	
95°C for 10 min	1 cycle
95°C for 15 sec	40 cycles
60°C for 1 min	

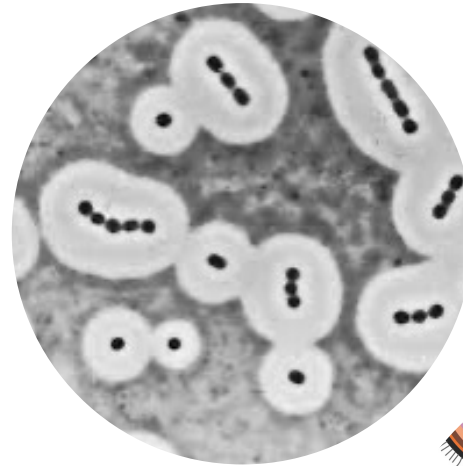
Note: Real-time PCR assays detecting: *lytA_sodC_hpd*

[illegible]

Practical 2 – *N. meningitidis* genogrouping

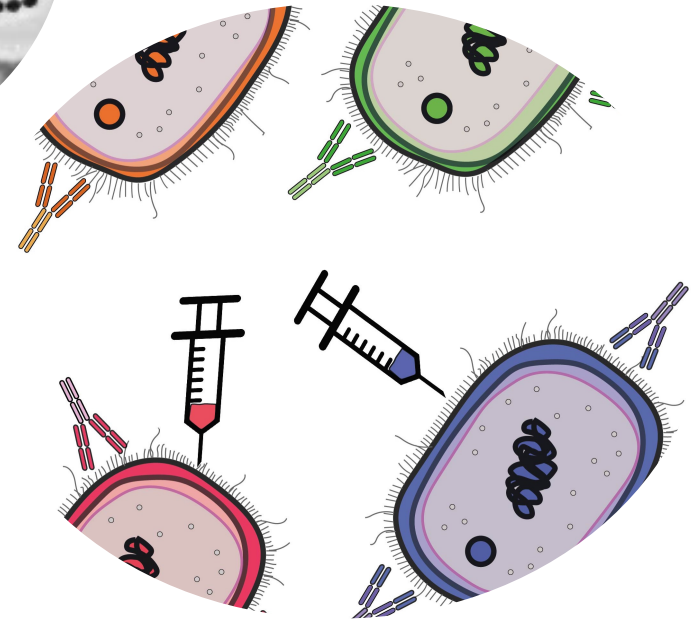
What is a capsule?

- Composed of polysaccharides;
- Protects bacteria from phagocytosis;
- Promotes adherence
- Prevents desiccation



Why are they important?

- They can be immunogenic;
- Important components of several polysaccharide-conjugate vaccines



	x1 (µl)	x4 (µl)
Reagents		
Water	19.75	79
Buffer mix	16	64
NmA F	1	4
NmA R	1	4
NmB F	1	4
NmB R	1	4
NmC F	1	4
NmC R	1	4
NmW F	1	4
NmW R	1	4
NmX F	1	4
NmX R	1	4
NmY F	1	4
NmY R	1	4
Taq	0.25	1



<i>Neisseria meningitidis</i> serogrouping												
	1	2	3	4	5	6	7	8	9	10	11	12
A	CG1NTC	CG2NTC	CG3NTC	CG4NTC	CG5NTC	CG6NTC	CG7NTC	CG8NTC	CG9NTC	CG10NTC	CG11NTC	CG12NTC
B	CG1S1	CG2S1	CG3S1	CG4S1	CG5S1	CG6S1	CG7S1	CG8S1	CG9S1	CG10S1	CG11S1	CG12S1
C	CG1S2	CG2S2	CG3S2	CG4S2	CG5S2	CG6S2	CG7S2	CG8S2	CG9S2	CG10S2	CG11S2	CG12S2
D											NmA ctrl	NmW ctrl
E											NmB ctrl	NmX ctrl
F											NmC ctrl	NmY ctrl
G												
H												

94C / 5 minutes



94C / 30 seconds

55C / 30 seconds



35 cycles



72C / 10 minutes



4C

